

Final Consulting Project Report

Spanish Intonation Analysis

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Contents

1	Summary	3
2	Project Objectives	3
2.1	Problem Statement	3
2.2	Project Objectives	3
3	Deliverables	4
4	Data	4
5	Methods	5
6	Findings	5
7	Future Improvements	6
A	Appendix	6
A.1	Exploratory visualizations	6
A.1.1	Overall response distributions	7
A.1.2	Sentence-type visualizations	8
A.1.3	Speaker-specific response profiles	11
A.2	Bootstrap confidence intervals	13

1 Summary

This project studies whether Ecuadorian speakers who lived in Spain and later returned to Ecuador still produce intonation patterns associated with Peninsular Spanish, whether they revert to Ecuadorian Spanish patterns, or whether they show intermediate or other distinct patterns. The data consist of repeated sentence-level recordings from six returnee speakers, along with speaker characteristics such as age, gender, education, length of stay in Spain, and time since return to Ecuador.

Because the study includes only six speakers but many sentence types and repeated observations, the main challenge was severe data sparsity, which made the originally intended mixed-effects modeling approach computationally unstable. The analysis therefore began with descriptive exploration and confidence-interval-based summaries, then moved to a simplified multinomial fixed-effects model after collapsing both the response categories and the sentence-type groups so that the research questions could still be addressed in a statistically workable way.

2 Project Objectives

2.1 Problem Statement

The central problem in this project is to determine whether Ecuadorian returnees who lived in Spain continue to produce intonational patterns associated with Peninsular Spanish after returning to Ecuador, or whether their speech aligns more closely with Ecuadorian Spanish or other hybrid patterns. More broadly, the analysis seeks to identify which speaker-level and sentence-level factors are associated with the production of these different intonational categories.

2.2 Project Objectives

The objectives of this project were:

- Determine whether Ecuadorian returnees who previously lived in Spain produce intonational patterns associated with Peninsular Spanish after returning to Ecuador.
- Assess whether these speakers instead produce patterns more consistent with Ecuadorian Spanish, or whether they exhibit intermediate or other non-canonical patterns.
- Compare the relative frequency of Ecuadorian, Spanish, and other intonational patterns across speakers and sentence types.
- Evaluate whether speaker-level characteristics such as age, gender, education, length of stay in Spain, and time since return to Ecuador are associated with the production of different intonational patterns.

- Examine whether sentence type influences the probability of producing Ecuadorian, Spanish, or other intonational categories.
- Develop a statistically workable modeling strategy for this sparse repeated-measures dataset that can support preliminary inference and be extended when more data become available.

3 Deliverables

The project deliverables included a written report, reproducible analysis code, and supporting explanation of the statistical methods so that the client could understand, replicate, and extend the work. In addition to producing the formal analyses, the project emphasized clear communication of interpreting and testing models and using the model(s) to make inferences in a way that supported practical decision-making and future use by the client.

4 Data

The analysis uses a speaker-level repeated-measures dataset designed to examine how returnee speakers produce Spanish patterns across different linguistic contexts. Each row in the dataset represents a single observed production and links that production to both speaker characteristics and sentence context.

The dataset includes six speakers and repeated observations across 17 sentence types. For each observation, the available variables include speaker identifier, age, gender, education level, time spent in Spain, time since return to Ecuador, sentence type, and a response-category outcome indicating the type of pattern produced.

The response variable was originally coded into five categories: **Ecuadorian**, **Spanish**, **Shared**, **Different**, and **Other**. Because the data are sparse at the speaker-by-sentence-type level, later stages of the analysis use a reduced response classification with three categories: **Ecuador**, **Spain**, and **Different**. This reduction improves interpretability and helps support more stable model estimation.

The sentence-type variable was also simplified for modeling purposes. The original 17 sentence types were grouped into four broader categories based on functional similarity: statements (Types 1–6), yes/no questions (Types 7–11), wh-questions (Types 12–14), and commands or other utterances (Types 15–17). This grouping was adopted to reduce sparsity and make cross-context comparisons more practical.

Several speaker-level predictors were treated as explanatory variables throughout the analysis. These included age, gender, education level, years spent in Spain, and years since return to Ecuador. Together, these variables were used to assess whether social and migration-related characteristics were associated with the likelihood of producing Spain, Ecuador, or other response patterns.

5 Methods

The analysis combined data cleaning, descriptive summaries, visualization, and multinomial modeling. Initial preprocessing included checking coding consistency, addressing minor data issues, and collapsing the original sentence-type and response variables into broader categories to reduce sparsity.

Descriptive statistics were used to summarize the distribution of responses across speakers and sentence contexts. Response-profile plots were then used to visualize category proportions by speaker and sentence type, with Wilson confidence intervals included to summarize uncertainty around observed proportions.

The main inferential analysis used a fixed-effects multinomial logistic regression model fit to the reduced three-category response outcome. Predictor variables included sentence group, age, gender, education level, years spent in Spain, and years since return to Ecuador. Predicted probabilities were computed from the fitted model to summarize how response patterns changed across covariates. Bootstrap confidence intervals were used for selected predicted probabilities to quantify estimation uncertainty.

A mixed-effects multinomial model was also considered to account for repeated observations within speakers. However, this approach was not retained in the final analysis because the data structure was too sparse to support stable estimation.

6 Findings

The descriptive summaries, response-profile plots, and multinomial model all suggested that speakers did produce Spanish-type patterns, but these patterns did not dominate the data overall. Model-based predicted probabilities indicated that Ecuadorian patterns remained common, while Spanish-pattern probabilities were present but more limited in magnitude.

Response behavior varied across sentence groups. The model and accompanying predicted probabilities suggested that some sentence contexts were more likely than others to be associated with Spanish-pattern responses, indicating that sentence type remained an important source of variation in production.

The results also suggested that time spent in Spain and time since return to Ecuador were related to the probability of producing Spanish-type patterns. In contrast, age, gender, and education appeared less consistently influential in the fitted model, although these conclusions should be treated cautiously because of the limited sample size.

The category labeled **Different** remained substantively important throughout the analysis. This suggests that some responses did not align cleanly with either the Spanish or Ecuadorian patterns and may reflect intermediate forms, mixed realizations, or other response types not fully captured by the main binary contrast.

Speaker-level response-profile plots showed substantial heterogeneity across individuals. Some speakers showed more frequent Spanish-type responses in particular sentence contexts, whereas others showed stronger Ecuadorian or mixed response patterns, indicating that the observed variation was not uniform across speakers.

The analysis also identified important data and design limitations. The dataset was sparse after stratifying by speaker and sentence type, some speaker-level variables appeared internally inconsistent, and a mixed-effects multinomial model could not be fit reliably under the available sample structure. As a result, the findings should be interpreted as exploratory and primarily useful for identifying broad patterns rather than making strong population-level claims.

7 Future Improvements

The most important next step is to collect more data by including additional speakers. Because the current dataset is sparse, the priority should be not only a larger sample, but a more representative one, with better coverage across speakers, sentence contexts, and other relevant covariates.

A second improvement would be to adopt a Bayesian modeling framework. This would better match the goals of the study, which are centered on uncertainty quantification and evidence about competing response patterns rather than prediction alone.

A Bayesian approach may also be more effective for sparse data and repeated observations within speakers, particularly through hierarchical modeling and partial pooling. These methods could produce more stable estimates and more interpretable uncertainty summaries than the current fixed-effects approach.

Future work should also include stronger data validation before analysis. Improving consistency in speaker-level records and clarifying aspects of the study design would support more reliable inference and allow for richer models.

A Appendix

A.1 Exploratory visualizations

This appendix collects the main exploratory plots and bootstrap confidence-interval summaries used to address the research questions. The visualizations were included to summarize response distributions across speakers, sentence types, and response categories, while the bootstrap tables were used to quantify uncertainty in model-based predicted probabilities.

A.1.1 Overall response distributions

Figure 1 shows the overall distribution of response categories in the full dataset. This visualization was used to summarize the baseline composition of the observed responses before fitting the multinomial model.

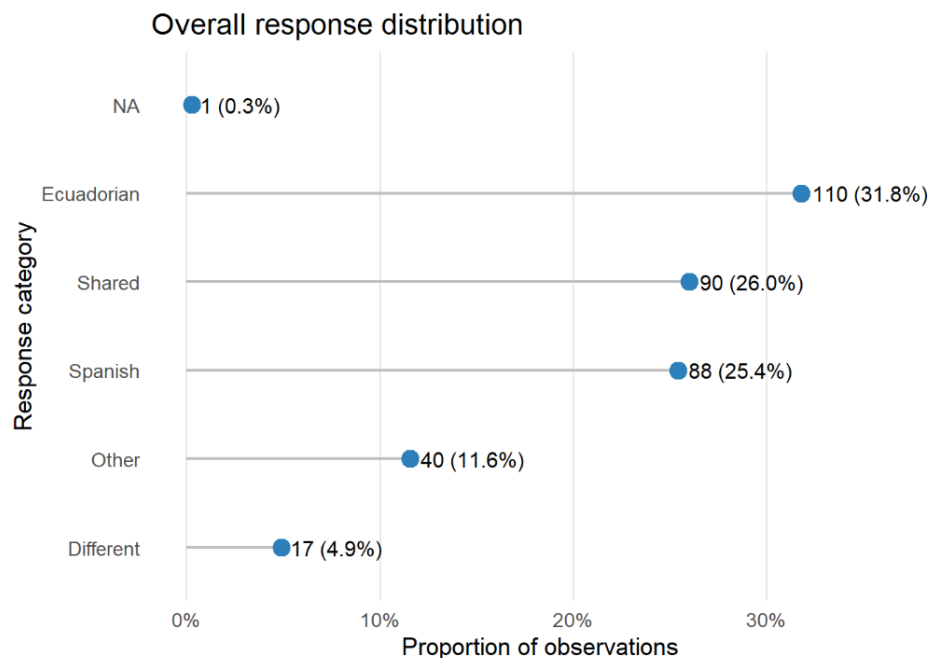


Figure 1: Overall response distribution across all observations.

Figure 2 shows the distribution of response categories by speaker. This plot was used to assess heterogeneity across speakers and to identify whether some speakers showed stronger tendencies toward particular response types.

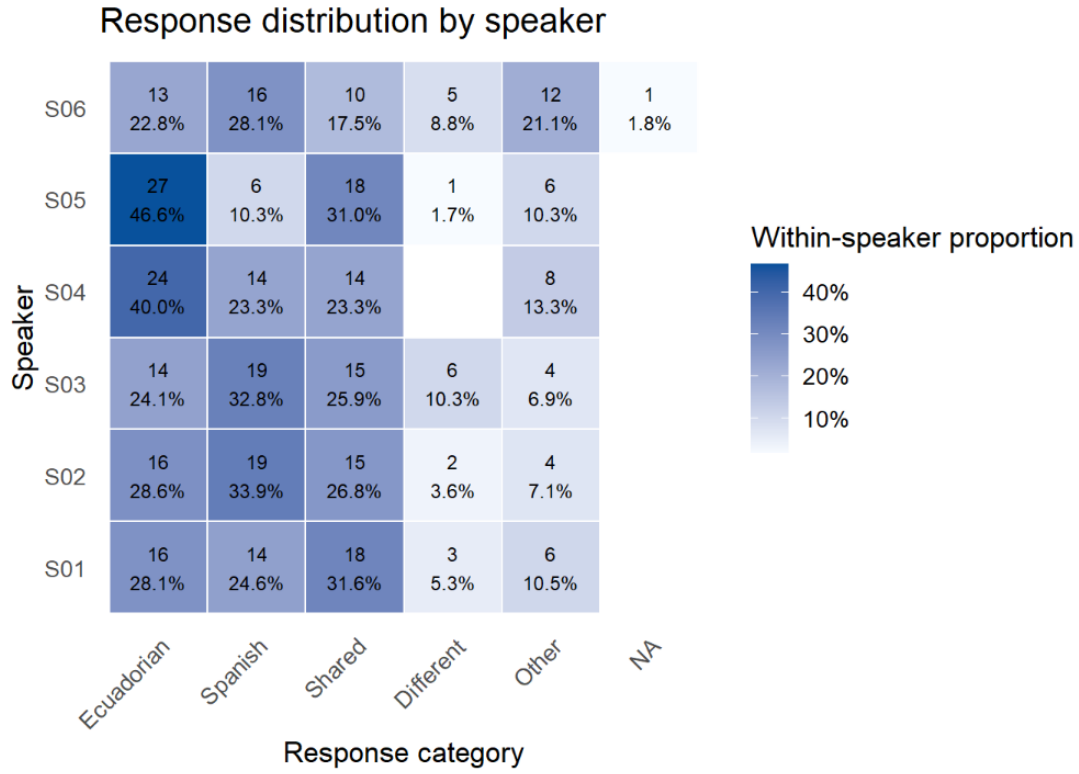


Figure 2: Response distribution by speaker.

A.1.2 Sentence-type visualizations

Figures 3–6 summarize the response distribution by sentence type, grouped into blocks for readability. These plots were used to examine whether response behavior differed across sentence contexts.

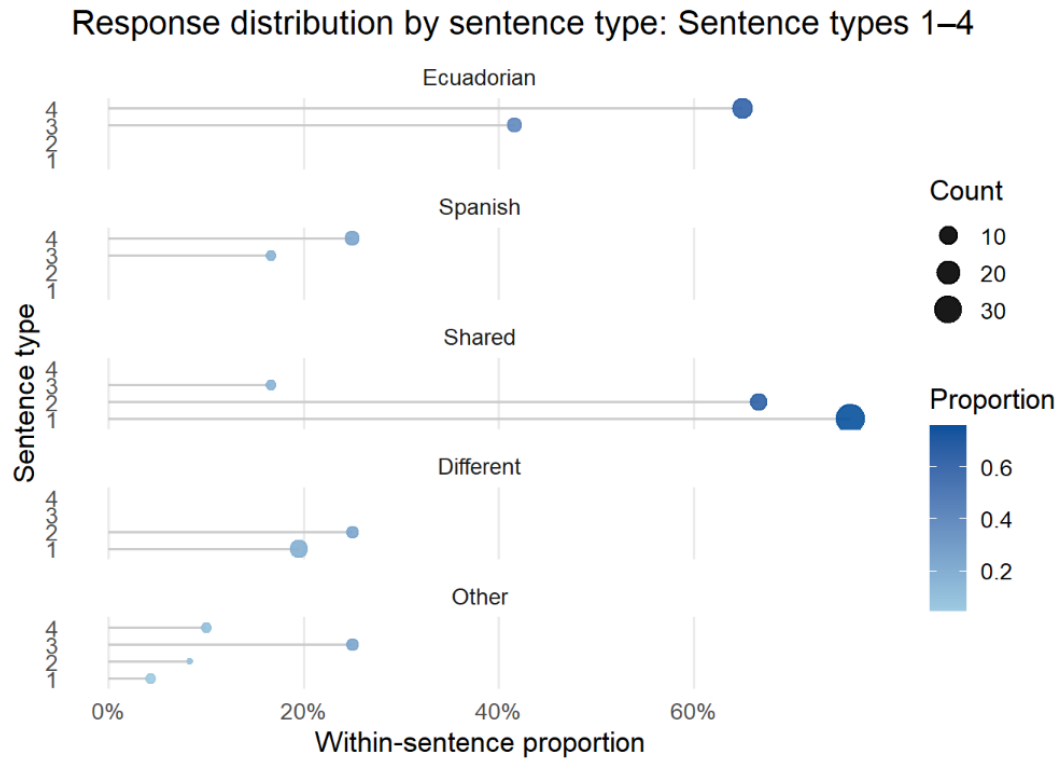


Figure 3: Response distribution by sentence type: sentence types 1–4.

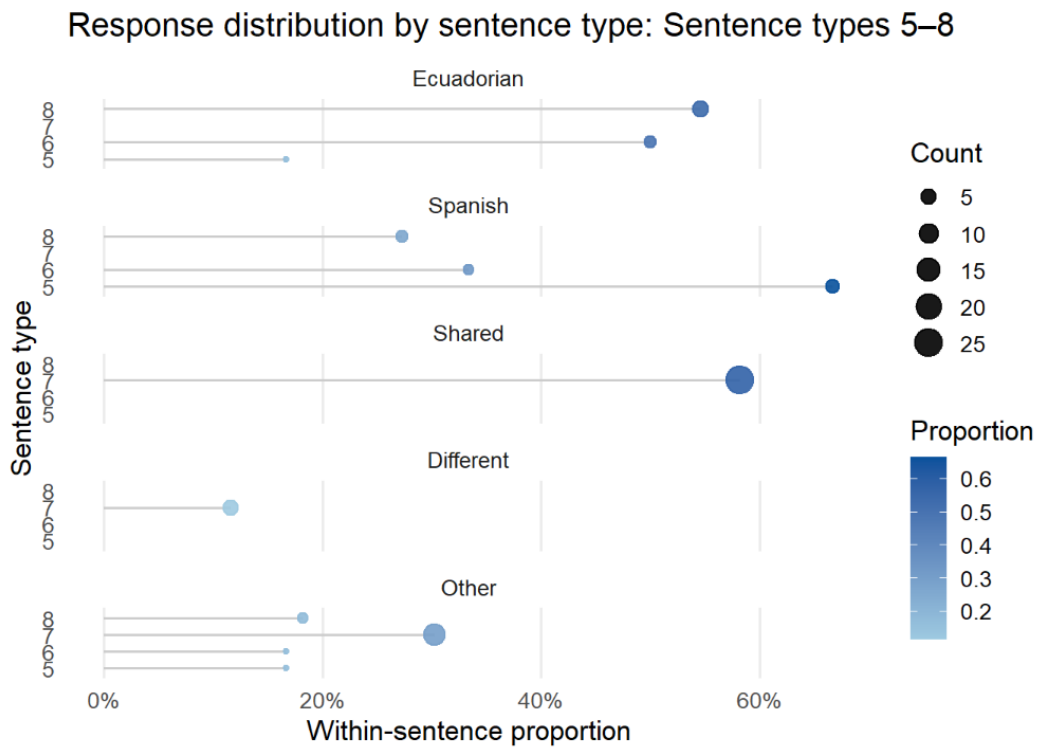


Figure 4: Response distribution by sentence type: sentence types 5–8.

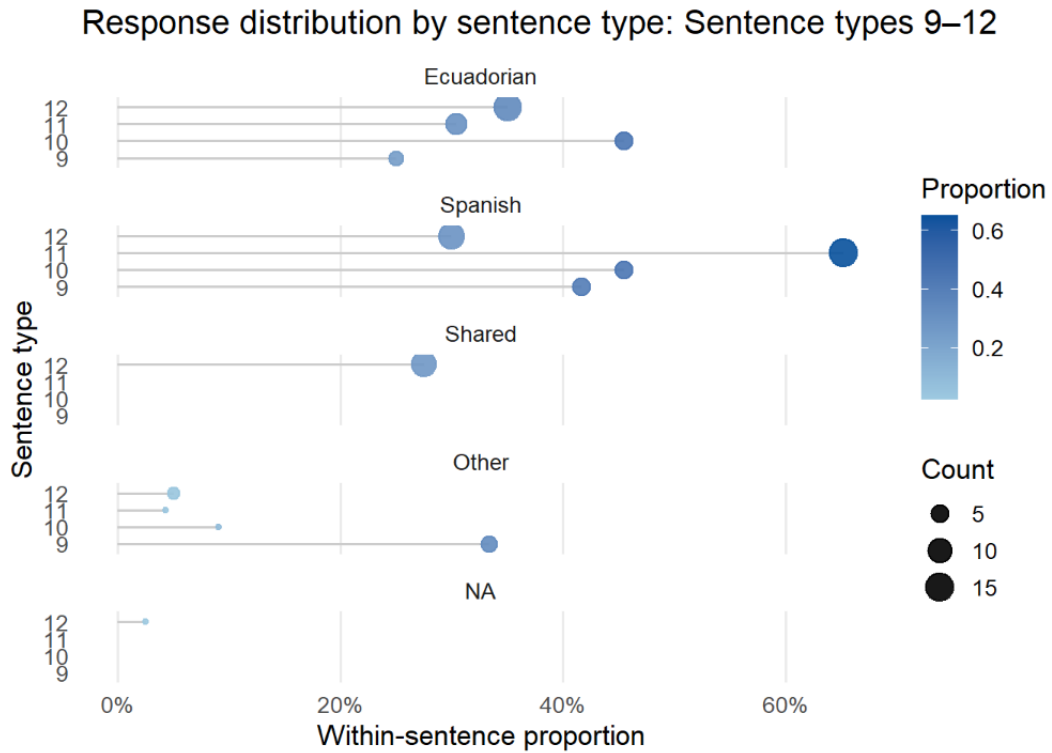


Figure 5: Response distribution by sentence type: sentence types 9–12.

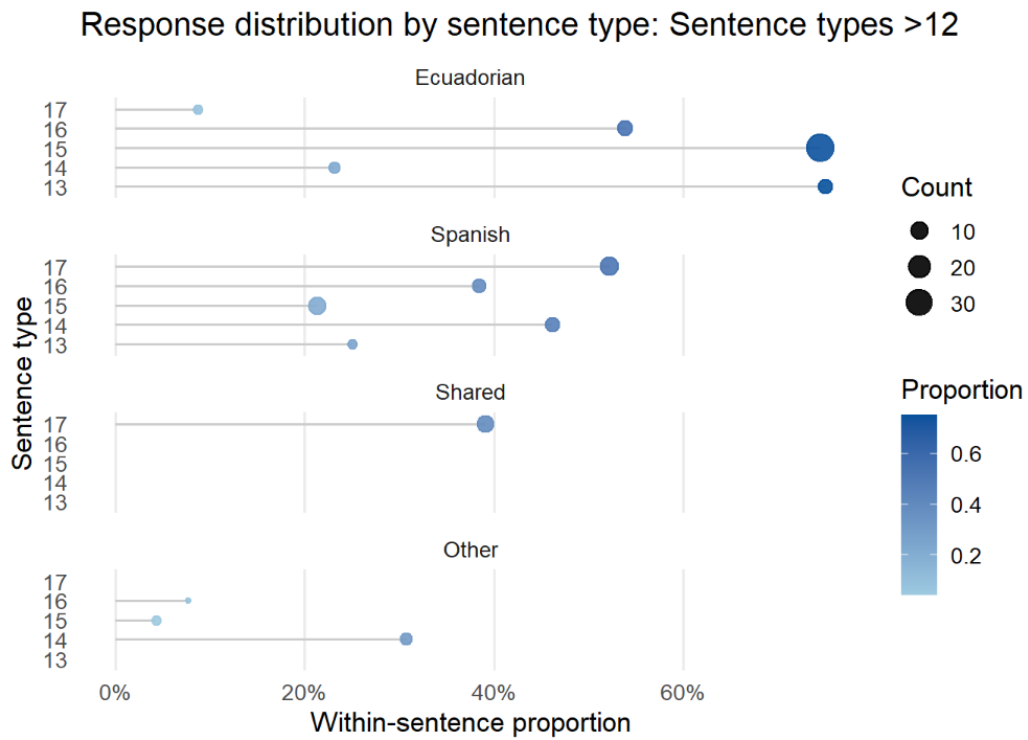


Figure 6: Response distribution by sentence type: sentence types greater than 12.

Figure 7 shows the distribution of sentence types across speakers. This visualization was used primarily to assess imbalance and sparsity in the observed design.

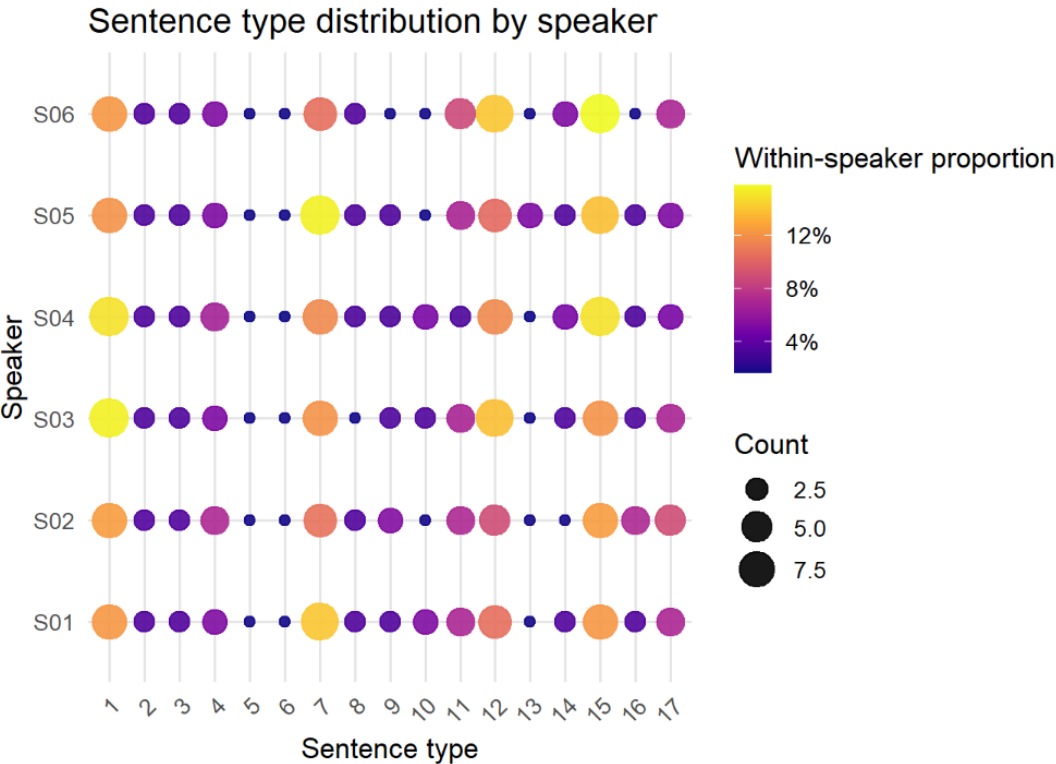


Figure 7: Sentence-type distribution by speaker.

A.1.3 Speaker-specific response profiles

Speaker-specific response-profile plots were used to visualize the distribution of response categories across sentence types within each speaker. Wilson confidence intervals were added to these plots to summarize uncertainty in observed proportions for small cell counts.



Figure 8: Response profile for speaker S01.

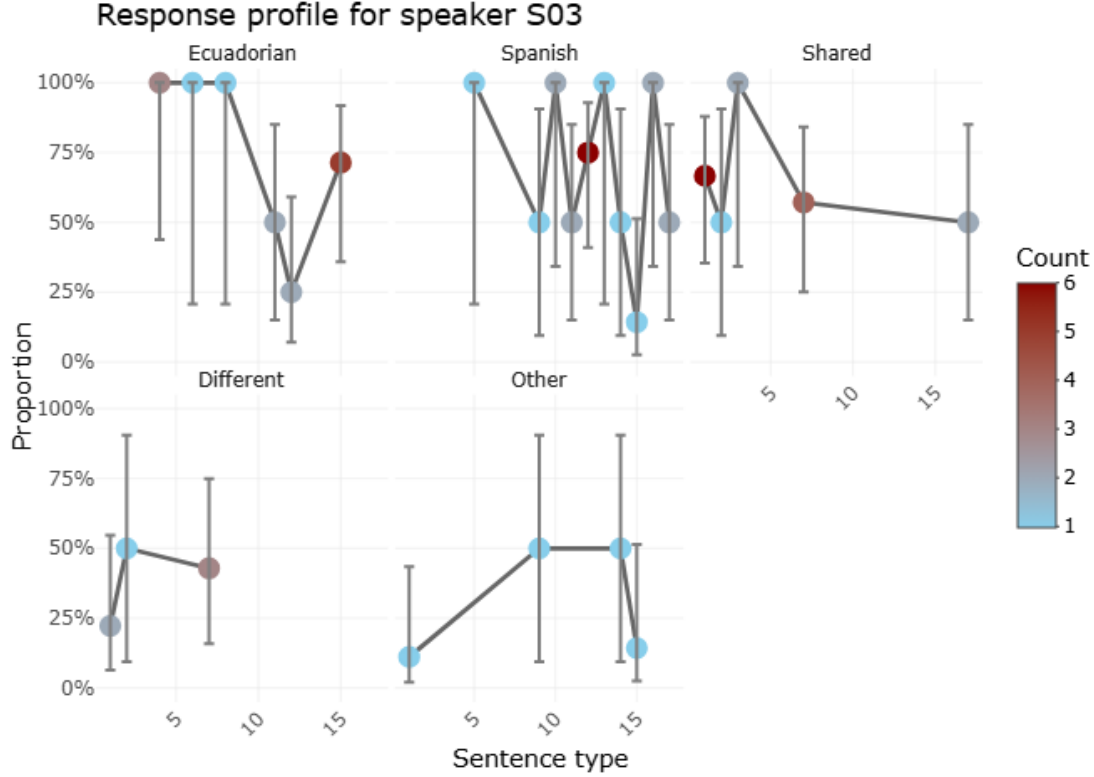


Figure 9: Response profile for speaker S03.

Additional speaker-specific profile plots may be included here as needed for the remaining speakers.

A.2 Bootstrap confidence intervals

The following tables report bootstrap confidence intervals for model-based predicted probabilities of Spanish-pattern responses. These tables were used to address the research questions concerning whether Spanish patterns were produced, under which conditions they were more likely, and how uncertainty in those probabilities should be summarized.

Table 1 reports the overall predicted probability of Spanish-pattern responses together with its bootstrap confidence interval. This table addresses whether Spanish patterns were produced in the sample and provides a corresponding uncertainty summary.

Table 1: Bootstrap confidence interval for the overall predicted probability of Spanish-pattern responses.

Quantity	Estimate	Lower 95% CI	Upper 95% CI
Overall Spanish probability	0.2558	0.2093	0.3023

Table 2 reports bootstrap confidence intervals for the predicted probability of Spanish-pattern responses by sentence group. This table addresses how sentence context relates to the probability of producing Spanish-type patterns.

Table 2: Bootstrap confidence intervals for Spanish-pattern probability by sentence group.

Sentence group	Estimate	Lower 95% CI	Upper 95% CI
Statement	0.1262	0.0718	0.1934
Yes/no question	0.2837	0.1940	0.3779
Wh-question	0.3419	0.2218	0.4476
Command/other	0.3212	0.2252	0.4198

Table 3 reports bootstrap confidence intervals for the predicted probability of Spanish-pattern responses for the observed combinations of time spent in Spain and time since return to Ecuador. This table addresses whether migration-related variables appear to be associated with Spanish-pattern production.

Table 3: Bootstrap confidence intervals for Spanish-pattern probability by observed combinations of time in Spain and time since return to Ecuador.

Stay in Spain	Return to Ecuador	Estimate	Lower 95% CI	Upper 95% CI
16	0.4167	0.2038	0.1109	0.2970
12	0.0493	0.3061	0.2463	0.3752
5	0.1667	0.2455	0.1366	0.3978
11	0.0274	0.3108	0.2396	0.3803
12	0.6667	0.1405	0.0603	0.2404
13	0.0575	0.3058	0.2435	0.3810